Gregorius Karisma

Data & Industry Analytics | Technical Project Lead | Transforming Multivariable Engineering & Enterprise Data into Scalable Decisions | MS-ISM @ ASU | Python • SQL • Agile

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SUMMARY

Data intelligence and systems architecture specialist with an academic background spanning an M.S. in Information Systems Management and previously a 3.92 GPA in Applied Business & Technology / Engineering. Skilled in transforming dense, multivariable engineering datasets and disparate enterprise records into scalable business intelligence and resilient software architectures. Proven track record extracting high-conviction signals using Python, SQL, R, and Tableau, delivering sustainability optimization models for Intel, digital strategy analytics for The Recording Academy, and enterprise-grade computational engines in C#/.NET. Adept in Agile product ownership, cross-functional engineering governance, and adapting quantitative physics with strategic executive decision-making.

CORE COMPETENCIES

Data Intelligence & Modeling: SQL, Python, R, Tableau, Microsoft Excel (Advanced Modeling), Predictive Modeling, Exploratory Data Analysis (EDA), Statistical Forecasting, Design of Experiments (DOE), Quantitative Analysis, Dark Data Structuring & Pipeline Architecture.
Systems Architecture & Engineering: C#, .NET 10.0, Windows Presentation Foundation (WPF), Object-Oriented Programming (OOP), Inter-Process Communication (IPC/JSON), Semiconductor Device Physics, Advanced Packaging (2.5D/3D), Materials Informatics, Numerical Simulation, Electronic Design Automation (EDA).
Agile Governance & Leadership: Agile Technical Product Ownership, Scrum Framework, Jira, Confluence, MoSCoW Prioritization, Technical Roadmapping, Code Review Governance, Cross-Functional Team Leadership, Stakeholder Reporting.
Languages: English (Native/Bilingual), Mandarin (Full Professional Proficiency — Simplified & Traditional), Taiwanese (Elementary).

EXPERIENCE

Technical Project Lead Nov 2025 – Present
GK & Team Tempe, Arizona

- Direct organization software architecture and technical roadmaps, leading developer teams to deliver full-cycle software deployments with a crash-free user experience.
- Operationalize various project requirements into actionable engineering specifications, collaborating cross-functionally to deliver fully defined functional modules.
- Ensure functional application quality by assigning technical tasks, conducting in-depth code reviews, and enforcing industry-standard design principles.
- Standardize team coding practices by sharing technical expertise in mathematical theories, physical parameters, and user control, significantly reducing bug reports.
- Drive on-time milestone completion across active sprints, resolving technical bottlenecks to maintain consistent project momentum.

Research and Development Analyst Feb 2025 – Oct 2025
GK & Team Tempe, Arizona

- Led technology intelligence initiatives, conducting in-depth market research and analyzing academic literature alongside aggregated datasets to establish project baselines.
- Built a centralized research repository and structured methodology database, saving significant research time across cross-functional engineering teams.
- Prototyped innovative technical solutions through structured brainstorming and executed full-scale feasibility testing, increasing overall project adoption rates.
- Formulated and delivered data-driven research reports to student leadership, directly shaping strategic technical decisions and project direction.

Participant – Global Career Accelerator: Data Analytics Jan 2025 – May 2025
Arizona State University Tempe, Arizona

- Mastered the complete data analytics lifecycle, executing end-to-end data extraction, transformation, cleaning, and visual storytelling across multidimensional datasets.

- Developed optimized analytical code and database queries utilizing SQL, Python, Microsoft Excel formulas, and Tableau dashboards.
- Built high-accuracy predictive models for various business scenarios, uncovering actionable operational trends through applied statistical methods.
- Communicated analytical findings and data-driven recommendations to stakeholder panels via structured executive presentations.

PROJECTS

Bachelor's Capstone: Thermal-Mechanical Stress Simulator for 2.5D/3D Semiconductor Packaging Aug 2025 – Jun 2026
- Architected a custom desktop simulation program evaluating thermal and mechanical stress responses under extreme power densities projected for 2026–2031 packaging.
- Evaluated critical performance metrics (maximum temperature gradients, shear stresses, and delamination risks) comparing Aluminum Nitride ceramics against Silica-Filled Epoxies.
- Implemented a decoupled dual-language architecture connecting a responsive C# WPF graphical interface to an isolated Python virtual environment backend via JSON-based inter-process communication.
- Integrated an automated zero-footprint garbage collection protocol to purge temporary cache and simulation artifacts, optimizing host system memory.

Intel Project: Corporate Sustainability & Energy Data Analytics Mar 2025 – May 2025
- Utilized Intel's corporate data provided by their team for environmental sustainability to analyze enterprise energy consumption metrics across United States data centers and business operations.
- Cleaned and structured multidimensional operational data in Excel and designed interactive Tableau dashboards to monitor environmental key performance indicators.
- Delivered strategic, data-driven recommendations optimizing facility power utilization and advancing enterprise corporate sustainability targets.

The GRAMMYs Project: Website Audience Engagement Analytics Jan 2025 – Mar 2025
- Analyzed enterprise visitor logs and engagement datasets for Grammy.com (The Recording Academy) to identify behavioral patterns and content efficacy.
- Processed and transformed raw audience traffic data in Microsoft Excel and developed Tableau visual analytics highlighting high-performing editorial verticals.
- Presented digital strategy findings to stakeholders, providing data-backed recommendations to maximize web audience acquisition and retention.

MSE Comprehensive: Materials Science Computational Suite Aug 2022 – Aug 2026
- Engineered a native Windows desktop suite in C# targeting .NET 10.0 and WPF, implementing a Model-View-ViewModel (MVVM) architecture with 10 specialized calculation modules.
- Automated various mathematical routines including non-steady-state diffusion (Fick's Second Law), Fox equation glass transition modeling, and Bragg's Law X-ray diffraction patterns.
- Deployed the software as a standalone local executable with custom XAML styling and sophisticated exception handling, enabling zero-latency computation in restricted lab environments.

CIS 311: Agile Banking Customer Dashboard Mar 2026 – May 2026
- Served as Agile Product Owner directing a 5-member team, managing sprint backlogs and prioritizing features using the MoSCoW method across two Scrum sprints.
- Tracked development milestones, sprint velocity, and burndown charts in Jira and Confluence, successfully delivering an intuitive multi-module customer dashboard (867/905 points).

EDUCATION

Arizona State University Tempe, Arizona
Master of Science in Information Systems Management 2026 – 2027
Bachelor's in Applied Business & Technology / Engineering Aug 2024 – Aug 2026
- **Academic Standing:** GPA: 3.92 / 4.00 | Dean's List: Outstanding Academic Achievement (Multiple Semesters)
- **Technical Certificates:** Applied Business Data Analytics; Materials Science for Advanced Technological Applications
- **Relevant Coursework:** Single Variable & Vector Calculus, Semiconductor Devices & Processes 1–2, Data Analytics & Science (Python, Java, R, SQL), Modern Statistics for Data-Driven Decisions, Dark Data Migration and Architecture, Design of Experiments, Chip-Based VLSI Design

California State University, Fullerton Fullerton, California
General Studies (Undergraduate Coursework) Aug 2022 – May 2024
- Coursework: College Algebra & Pre-Calculus, Principles of Microeconomics & Macroeconomics, Financial Accounting, Introduction to Logic

PUBLICATIONS

Analysis of the Electric Field and Potential in a p-n Junction Using Gauss's Law Published Jul 2025
Arizona State University / Academia.edu
- Derived internal electrostatics of semiconductor p-n junctions using Gauss's Law and depletion approximations, modeling triangular electric fields and quadratic potential profiles.
- Formulated analytical expressions for built-in potential (V_{bi}), directly linking semiconductor doping parameters to rectifying diode performance.

LICENSES & CERTIFICATIONS

- **Google Data Analytics Professional Certificate**, *Google* (Credential ID: OWFDX73O23UN) Jan 2026
- **Introduction to Chip Design with Open-Source EDA Tools**, *Coursera* (Credential ID: N3FHP3K9963Z) Jun 2026

PROFESSIONAL AFFILIATIONS

North America Taiwanese Engineering & Science Association (NATEA) Sep 2026 – Present
Member, Arizona Chapter (NATEA-AZ) Phoenix / Tempe, Arizona
- Engage with regional semiconductor and advanced manufacturing industry networks, facilitating cross-border technical collaboration between Arizona technology hubs and Taiwan high-tech ecosystems.
- Participate in technical symposiums, startup showcases, and venture forums evaluating emerging hardware engineering, advanced packaging architectures, and materials science innovations.

VOLUNTEERING

Student Organizations: Google Developer Student Club at ASU (Member, Nov 2024 – Sep 2026) • Taiwanese Students Association (Member, Nov 2025 – Present) • Singapore Student Association (Member, Aug 2024 – Sep 2026) • Model United Nations (Jul 2019 – May 2021)

Field Operations Coordinator, *Community Service Program (Greater Jakarta, Indonesia)* Oct 2020 – Feb 2021
- Managed distribution logistics and inventory tracking for 40 hours of field operations, directing essential aid deliveries (clean water, food, medicine, and educational supplies) to rural communities.